KELCEY DAVIS

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LinkedIn ◇ Personal Website

PROFESSIONAL APPOINTMENTS

Visiting Student Los Alamos National Lab, X Theoretical Design (XTD) — Fall 2025-Present
Intern Los Alamos National Lab, X Computational Physics (XCP) — Summer 2025
NSF Graduate Fellow The University of Connecticut — August 2022 - Present
NSF Graduate Fellow Brown University — August 2021 - August 2022
MARC Pre-Doctoral Fellow San Diego State University — September 2019-May 2021
Cal-Bridge Pre-Doctoral Fellow San Diego State University — September 2019-May 2021
CHAMP REU Student University of Washington — Summer 2019

EDUCATION

University of Connecticut (UConn) — *Expected May 2027*
PhD Candidate in Physics
San Diego State University (SDSU) — May 2021
Bachelor of Science in Astronomy, Awarded Outstanding Undergraduate
Double Minors in Mathematics and Creative Writing
San Diego City College — May 2019
Associate of Science in Astronomy

AWARDED GRANTS (PI), FELLOWSHIPS, AND RECOGNITIONS

- **PI: Davis, K.**, co-PI: Larson, R. et al. 2025 " SPAM: Star-formation from Photometry through the Addition of Medium-bands " JWST Proposal. Cycle 4, ID. #8559, **62.8 Primary Science Hours, $354,520**
- **NSF Graduate Research Fellowship** (NSF GRFP), 2021
- **Outstanding Undergraduate**, 2021 Honor Awarded to a single graduating senior anuually by Department of Astronomy

AWARDED GRANTS (CO-I)

- PI: Cleri, Nikko ...**co-I: Davis, K.**, et al. 2026 " CEDAR: Characterizing Electron Densities At Reionization" JWST Proposal. Cycle 5, ID. #10518, **29.2 Primary Science Hours**
- PI: Marshall, Madeline ...**co-I: Davis, K.**, et al. 2026 " JWST observations of high-z quasar environments with the BlueTides simulation" JWST Proposal. Cycle 5, ID. #10799, **Archival**
- PI: Simons, Raymond ...**co-I: Davis, K.**, et al. 2025 " A Census of Galaxy Kinematics and Outflows to z $\sim$ 7 (Renamed the OCEANS Collaboration) " JWST Proposal. Cycle 4, ID. #8410, **65.3 Primary Science Hours**
- PI: Cleri, Nikko J., ...**co-I: Davis, K.**, et al. 2024 "A Census of Optical Diagnostics of Ionizing Sources Across Cosmic Time " JWST Proposal. Cycle 3, ID. #5558, **Archival**
- PI: Kocevski, Dale D., ...**co-I: Davis, K.**, et al. 2024 "A Spectroscopic Census of Faint, Broad-Line AGN at $z > 5$ " JWST Proposal. Cycle 3, ID. #5718, **20.49 Primary Science Hours**

COLLABORATION MEMBERSHIP

Star-formation from Photometry through the Addition of Medium-bands (SPAM) Summer 2026 - present PI: **Kelcey Davis**, JWST NIRCam photometry. Details on the SPAM website

Observing Cosmic Evolution Among Nascent Systems (OCEANS) Spring 2024 - present PI: Raymond Simons, JWST NIRSpec high-resolution spectroscopy

The High-(Redshift+Ionization) Line Search (THRILS) Spring 2024 - present PIs: Rebecca Larson, Taylor Hutchison, JWST deep NIRSpec spectroscopy

The CANDELS-Area Prism Epoch of Reionization Survey (CAPERS) Fall 2024 - present PI: Mark Dickinson, JWST NIRSpec prism spectroscopy

Cosmic Evolution Early Release Science (CEERS) Survey Fall 2022 - present, PI: Steve Finkelstein, JWST early release science program

The Hydrogen Epoch of Reionization Array (HERA) Summer 2019 - Summer 2022, PI: Aaron Parsons, 21cm radio telescope

RESEARCH PRESENTATIONS

- **19** oral presentations and seminars, **7** conference posters
- Selected venues: UMBRELA Harvard Center for Astrophysics Dialogue Series, Los Alamos Astrophysics Distinguished Seminar Series, and the New York Times Science Journalism Course

TEACHING AND OUTREACH

Details for all listed projects
Scientist, Data Communicator STEMArts Fall 2025 - Present
Planetarium Presenter Los Alamos Pajarito Environmental Education Center (PEEC) Summer 2025 - Present
Teaching Assistant, University of Connecticut, January 2025 - May 2025
Co-founder, Vice President BEAConn AI Safety Organization, Fall 2024 - Spring 2025
Planetarium Presenter University of Connecticut, Fall 2024 - Summer 2025
Research Mentor Boyce Astro October 2018 – August 2022

SKILLS

Programming Languages	Python, FORTRAN (90)
General Software	LaTeX, Unix, Git, Claude
Working Knowledge	IDL, HTML, Matlab
Statistics	χ^2 fitting, Bayesian Information Criteria, weighted least squares, quantifying statistically strong corelation (Spearman p-values), quantitative/qualitative modeling, error quantification, big data, verification and validation
Astronomy Skills	SED fitting (`EAZY`), photoionization modeling comparisons (MAPPINGS, `pyneb`), emission-line fitting (spectroscopic and photometric), photometric color comparisons, half-light radii/ compactness quantification, AGN identification, radiative rates and cross sections for particle interactions, plasma opacities
Observation Proposal and Planning	Astronomer's Proposing Tool (APT), Exposure Time Calculator (ETC), JWST NIRCam photometric survey design, JWST/NIRSpec MSA target selection across prism (R$\sim$ 100), medium-resolution grating (R$\sim$ 1000), and high-resolution grating (R$\sim$ 2700) modes; targeting and interpretation
Other skills	Managing and working in large international teams, mentoring junior researchers, grant writing, public speaking, science communication across skill level

RESEARCH MENTEES

8. Ian Bishop, the Institute for Computing in Research, Los Alamos National Lab, Summer 2026
7. Akshaj Vyas, the Institute for Computing in Research, Los Alamos National Lab, Summer 2026
6. Kate Bopp, the Institute for Computing in Research, Los Alamos National Lab, Summer 2026
5. Jessica Mitlehner, University of Connecticut January 2023 - May 2024
4. Mark S. Harris, Boyce Astro, January 2020 - December 2021
3. Jordan Guillory, Boyce Astro, October 2019 - January 2020
2. Velne Barud, Boyce Astro, October 2019 - January 2020
1. Neusha Kharrati, Boyce Astro, May 2019 - October 2019

JOURNAL PUBLICATIONS

Highlights Full record & citation statistics on SciX

2,300+ total citations and an **h-index of 19**; **i10-index of 22** and **i100-index of 8**

First Author Publications

1. **Davis, K.**, Larson, R., Akins, H., et al. 2026 "SPAM: Survey Overview and JWST/NIRCam multi-band Photometry Catalog" in preparation, expected October
2. **Davis, K.**, Brooks, M., Simons, R. C., et al. 2026 "OCEANS of Absorption: High-resolution NIRSpec Spectroscopy Reveals Diverse Balmer-line Absorption in Little Red Dots" submitted (arXiv:2606.00258)
3. **Davis, K.**, Brooks, M., Trump, J. R., et al. 2026 "Extreme Emission Line Galaxies in CEERS Are Powered by Star Formation, not AGN" submitted (arXiv:2602.23310)
4. **Davis, K.**, Trump, J. R., Simons, R. C. et al. 2024 "A Census from JWST of Extreme Emission-line Galaxies Spanning the Epoch of Reionization in CEERS" The Astrophysical Journal, Volume 974, Issue 1, id.42, 21 pp.

Second Author Publications

5. Brooks, M., **Davis, K.**, Trump, J. R., et al. 2026 "Ripples in the OCEANS: Broad Line Variability of Little Red Dots" submitted (arXiv:2608.12487)
6. Tran Tan, H. B., **Davis, K. S.**, Johnson, C. E., et al. 2026 "Electron-impact ionization from excited states method" High Energy Density Physics, Volume 58, id.101270

Supporting Author Publications

7. Ganapathy, Ananya, Larson, Rebecca L., Lambrides, Erini, ...**Davis, K.**, et al. 2026 "The THRILS Factor: Investigating the properties of Little Red Dots (LRDs) at $3 < z < 6$ with JWST/NIRSpec" submitted (arXiv:2609.10803)
8. Kocevski, Dale D., Taylor, Anthony J., Onoue, Masafusa, ... **Davis, K.**, et al. 2026 "Skyfire: A Spectroscopic Census of Little Red Dots and Broad-Line AGN in the CEERS Field" submitted (arXiv:2609.00112)
9. Johnson, Jarrett, Upton Sanderbeck, Phoebe, Lloyd-Ronning, Nicole, ...**Davis, K.** 2026 "The Lifetimes of High-redshift Quasars Suggest Magnetic Disk Support" The Astrophysical Journal, Volume 1006, id.84
10. Roy, Niranjan Chandra, Anglés-Alcázar, Daniel, Cochrane, Rachel K., ...**Davis, K.**, et al. 2026 "Little Red Dots on FIRE: Exploring the formation and observational signatures of ultra-compact early galaxies" submitted (arXiv:2606.23683)

11. Brooks, Madisyn, Trump, Jonathan R., Simons, Raymond C., ...**Davis, K.**, et al. 2026 "Beyond the Monsters: A More Complete Census of Black Hole Activity at Cosmic Dawn" The Astrophysical Journal, Volume 1002, id.129

12. Llerena, M., Pentericci, L., Amorín, R., ...**Davis, K.**, et al. 2026 "Extreme equivalent-width-selected low-mass starbursts at $z = 4-9$: Insights into their role in cosmic reionization" Astronomy & Astrophysics, Volume 708, id.A152

13. McGrath, Elizabeth J., Finkelstein, Steven L., Barro, Guillermo, ...**Davis, K.**, et al. 2026 "A Morphology Catalog of Galaxies in CEERS: Evolution in the Size and Color Gradients of Galaxies Since Cosmic Dawn" The Astrophysical Journal Letters, Volume 999, id.L6

14. White, J. R., Johnson, C. E., **Davis, K. S.**, et al. 2026 "Quantum fluctuations in dense plasma simulations" Physics of Plasmas, Volume 33, Issue 3, id.032701

15. Gentile, Fabrizio, Giavalisco, Mauro, Daddi, Emanuele, ...**Davis, K.**, et al. 2026 "The quasi-star model for Little Red Dots: potential and challenges" submitted (arXiv:2606.06575)

16. Pérez-González, Pablo G., Barro, Guillermo, Carniani, Stefano, ...**Davis, K.**, et al. 2026 "Little Red Dots: One Photometric Tag Concealing Diverse Spectroscopic Flavors of Massive Star Formation and Black Hole Activity" submitted (arXiv:2602.20247)

17. Mascia, S., Pentericci, L., Llerena, M., ...**Davis, K.**, et al. 2025 "Little impact of mergers and galaxy morphology on the production and escape of ionizing photons in the early Universe" Astronomy & Astrophysics, Volume 701, id.A122

18. Taylor, Anthony J., Kokorev, Vasily, Kocevski, Dale D., ...**Davis, K.**, et al. 2025 "CAPERS-LRD-z9: A Gas-enshrouded Little Red Dot Hosting a Broad-line Active Galactic Nucleus at z = 9.288" The Astrophysical Journal Letters, Volume 989, Issue 1, id.L7

19. Kokorev, Vasily, Chávez Ortiz, Óscar A., Taylor, Anthony J., ...**Davis, K.**, et al. 2025 "CAPERS Observations of Two UV-bright Galaxies at $z > 10$. More Evidence for Bursting Star Formation in the Early Universe" The Astrophysical Journal Letters, Volume 988, Issue 1, id.L10

20. Cleri, Nikko J., Olivier, Grace M., Backhaus, Bren E., ...**Davis, K.**, et al. 2025 "Optical Strong Line Ratios Cannot Distinguish Between Stellar Populations and Accreting Black Holes at High Ionization Parameters and Low Metallicities" The Astrophysical Journal, Volume 994, id.146

21. Morales, Alexa M., Finkelstein, Steven L., Arrabal Haro, Pablo, ...**Davis, K.**, et al. 2025 "Testing Photometric Techniques for Measuring the Rest-frame UV Spectral Slope against JWST PRISM Spectroscopy" The Astrophysical Journal, Volume 994, id.212

22. Brooks, Madisyn, Simons, Raymond C., Trump, Jonathan R., ...**Davis, K.**, et al. 2025 "Here There Be (Dusty) Monsters: High-redshift Active Galactic Nuclei Are Dustier than Their Hosts" The Astrophysical Journal, Volume 986, Issue 2, id.177

23. Taylor, Anthony J., Finkelstein, Steven L., Kocevski, Dale D., ...**Davis, K.**, et al. 2025 "Broad-line AGNs at $3.5 < z < 6$: The Black Hole Mass Function and a Connection with Little Red Dots" The Astrophysical Journal, Volume 986, Issue 2, id.165

24. Kocevski, Dale D., Finkelstein, Steven L., Barro, Guillermo, ...**Davis, K.**, et al. 2025 "The Rise of Faint, Red Active Galactic Nuclei at $z > 4$: A Sample of Little Red Dots in the JWST Extragalactic Legacy Fields" The Astrophysical Journal, Volume 986, Issue 2, id.126

25. Finkelstein, Steven L., Bagley, Micaela B., Arrabal Haro, Pablo, ...**Davis, K.**, et al. 2025 "The Cosmic Evolution Early Release Science Survey (CEERS)" The Astrophysical Journal Letters, Volume 983, Issue 1, id.L4

26. Backhaus, Bren E., Cleri, Nikko J., Trump, Jonathan R., ...**Davis, K.**, et al. 2025 "Emission-line Diagnostics at $z > 4$: [OIII]λ4364/Hγ versus [NeIII]λ3870/[OII]λ3730" The Astrophysical Journal, Volume 994, id.125

27. Hutchison, Taylor A., Larson, Rebecca L., Arrabal Haro, Pablo, ...**Davis, K.**, et al. 2025 "THRILS – The High-(Redshift+Ionization) Line Search: Program Description & Redshift Catalog" submitted (arXiv:2512.12509)

28. Merlin, E., Santini, P., Paris, D., ...**Davis, K.**, et al. 2024 "ASTRODEEP-JWST: NIRCam-HST multi-band photometry and redshifts for half a million sources in six extragalactic deep fields" Astronomy & Astrophysics, Volume 691, id.A240

29. Llerena, M., Amorín, R., Pentericci, L., ...**Davis, K.**, et al. 2024 "Physical properties of extreme emission-line galaxies at z $\sim$ 4–9 from the JWST CEERS survey" Astronomy & Astrophysics, Volume 691, id.A59

30. Napolitano, L., Pentericci, L., Santini, P., ...**Davis, K.**, et al. 2024 "Peering into cosmic reionization: Lyα visibility evolution from galaxies at $z = 4.5 - 8.5$ with JWST" Astronomy & Astrophysics, Volume 688, id.A106

31. Finkelstein, Steven L., Leung, Gene C. K., Bagley, Micaela B., ...**Davis, K.**, et al. 2024 "The Complete CEERS Early Universe Galaxy Sample: A Surprisingly Slow Evolution of the Space Density of Bright Galaxies at $z \sim$ 8.5–14.5" The Astrophysical Journal Letters, Volume 969, Issue 1, id.L2

32. Pirzkal, Nor, Rothberg, Barry, Papovich, Casey, ...**Davis, K.**, et al. 2024 "The Next Generation Deep Extragalactic Exploratory Public Near-infrared Slitless Survey Epoch 1 (NGDEEP-NISS1): Extragalactic Star-formation and Active Galactic Nuclei at $0.5 < z < 3.6$" The Astrophysical Journal, Volume 969, Issue 1, id.90

33. Bagley, Micaela B., Pirzkal, Nor, Finkelstein, Steven L., ...**Davis, K.**, et al. 2024 "The Next Generation Deep Extragalactic Exploratory Public (NGDEEP) Survey" The Astrophysical Journal Letters, Volume 965, Issue 1, id.L6

34. Barro, Guillermo, Pérez-González, Pablo G., Kocevski, Dale D., ...**Davis, K.**, et al. 2024 "Extremely Red Galaxies at z = 5–9 with MIRI and NIRSpec: Dusty Galaxies or Obscured Active Galactic Nuclei?" The Astrophysical Journal, Volume 963, Issue 2, id.128

35. Backhaus, Bren E., Trump, Jonathan R., Pirzkal, Nor, ...**Davis, K.**, et al. 2024 "CEERS Key Paper. VIII. Emission-line Ratios from NIRSpec and NIRCam Wide-Field Slitless Spectroscopy at $z > 2$" The Astrophysical Journal, Volume 962, Issue 2, id.195

36. Guo, Yuchen, Jogee, Shardha, Finkelstein, Steven L., ...**Davis, K.**, et al. 2023 "First Look at $z > 1$ Bars in the Rest-frame Near-infrared with JWST Early CEERS Imaging" The Astrophysical Journal Letters, Volume 945, Issue 1, id.L10

Non-refereed Student Publications *Double-star astrometry from community-college research program (Boyce Astro); co-authors include undergraduate mentees*

- Kharrati, N.K., **Davis, K.**, et al. 2019 "Astrometric Measurements and Historical Data Summary of WDS 05599-3703 (B 1044AD)" Journal of Double Star Observations
- **Davis, K.**, Vanslyke, S., Boyce, P., et al. 2019 "An Astrometric Measurement of WDS 16476-4708 AB and AC" Journal of Double Star Observations